

Sec. 8.1&8.5: Sample Spaces, Events, and Probability;
Random Variable, Probability Distribution, and Expected Value

1

A **random experiment** (or experiment) is a process or activity which produces a number of possible outcomes. The outcomes cannot be predicted with absolute certainty.

Example 1: Flip two coins and observe the possible outcomes of heads and tails.

Example 2: Select two marbles without replacement from a bag containing 1 white, 1 red and 2 green marbles.

Example 3: Roll two dice and observe the sum of the points on the top faces.

sample space - a list of all possible outcomes of the experiment.

event - a collection of outcomes

Example

Experiment: select a card from an ordinary deck of playing cards (no jokers).

The **sample space** consists of the 52 cards, 13 of each suit. We have 13 clubs, 13 spades, 13 hearts and 13 diamonds.

An example of a **simple event** is that the selected card is the two of clubs.

An example of a **compound event** is that the selected card is red (there are 26 red cards and so there are 26 simple events comprising the compound event).

Example

Experiment: Roll two dice

Sample space:

(1,1), (1,2), (1,3), (1,4), (1,5), (1,6)

(2,1), (2,2), (2,3), (2,4), (2,5), (2,6)

(3,1), (3,2), (3,3), (3,4), (3,5), (3,6)

(4,1), (4,2), (4,3), (4,4), (4,5), (4,6)

(5,1), (5,2), (5,3), (5,4), (5,5), (5,6)

(6,1), (6,2), (6,3), (6,4), (6,5), (6,6)

Probability

2 types of probability

- **Theoretical** - This uses basic assumptions and a reasoning process.
- **Empirical** - This type of probability is determined by experimentation. We assign probabilities to events based on the results of actual experiments.

If there are n outcomes in the sample space and all outcomes in the sample space are equally likely, then each has a theoretical probability of $\frac{1}{n}$. The theoretical probability of an event is the number of

outcomes in the event divided by the number of outcomes in the sample space, $P(E) = \frac{n(E)}{n(S)}$

Example

A bag contains 100 marbles in which 30 are red, 20 are blue, 15 are yellow, and 35 are white? You draw one marble from the bag. What is the probability that it is white? What is the probability that it is red or yellow?

Example

Find the probability of a sum of 7 when two dice are rolled.

If we conduct an experiment n times and event E occurs with frequency $f(E)$, then the ratio $f(E)/n$ is called the **relative frequency** or **approximate empirical probability** of the occurrence of event E in n trials. Empirical probability relies upon the long run relative frequency of an event.

Example

Out of the last 1000 statistics students, 150 of the students received an A. What is the empirical probability that a student receives an A?

$$\frac{150}{1000} = \frac{3}{20} = 15\%$$

Random Variables

A **random variable** is a function that assigns a numerical value to each simple event in a sample space S .

Examples:

1. the number of speeding tickets issued on a certain stretch of I 95 S.
2. the number of heads which appear when 4 dimes are tossed.
3. the number of passes completed in a game by a quarterback.

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3

The point of introducing random variables is to associate the simple events with **numbers**, with which we can calculate.

We transfer the probability assigned to elements or subsets of the sample space to numbers. This is called the **probability distribution** of the random variable X . It is defined as

$$p(x) = P(X = x)$$

Example

Let X = number of heads that show when tossing three coins.

Sample space: HHH, HHT, HTH, THH, HTT, THT, TTH, TTT

If you performed this experiment many times, how many heads would you expect on average?

x	$p(x)$
0	$\frac{1}{8}$
1	$\frac{3}{8}$
2	$\frac{3}{8}$
3	$\frac{1}{8}$

$$\frac{8}{8} = 1$$

The **expected value** of a random variable X is defined as $E(X) = x_1p(x_1) + x_2p(x_2) + \dots + x_n p(x_n)$

Example

Ten thousand raffle tickets are sold at \$2 each for a local library benefit. Prizes are awarded as follows: 2 prizes of \$1,000, 4 prizes of \$500, and 10 prizes of \$100. What is the expected value of this raffle if you purchase 1 ticket?

Example

The annual premium for a \$5,000 insurance policy against the theft of a painting is \$150. If the (empirical) probability that the painting will be stolen during the year is .01, what is your expected return from the insurance company if you take out this insurance?

$$E(x) = 5000(.01) + 0(0.99) - 150$$

Retn P(x)

5000	.01
0	.99

$$50 + 0 - 150 = -100$$

Review/Practice

1. Solve: $9(x - 8) = 7(x + 2) - 10$
2. Solve: $8(2x - 7) < 48$
3. Find the slope and y-intercept of $y = -4x + 7$. Sketch the graph.
4. Write an equation of the line with slope 12 that passes through the point $(5, -3)$.
5. If $f(x) = -x^2 + 4x + 8$, evaluate and simplify
 - a. $f(3)$
 - b. $f(-1)$
 - c. $f(4x)$
6. Find and simplify $\frac{f(x+h)-f(x)}{h}$ for
 - a. $f(x) = 9x - 2$
 - b. $f(x) = (2x+3)$ $2x^2+3$
7. Sketch the graphs of $f(x) = x$, $f(x) = x^2$, $f(x) = x^3$, $f(x) = \sqrt{x}$, $f(x) = \sqrt[3]{x}$, and $f(x) = |x|$.
8. Find the vertex of
 - a. $f(x) = (x + 2)^2 - 5$
 - b. $f(x) = 2x^2 - 8x + 7$